

Hannah Zheng

hannah_zheng@berkeley.edu | (510)-586-8398 | www.linkedin.com/in/hannah_zheng

EDUCATION

University of California, Berkeley

B.A Mechanical Engineering, Intended Minor in Bioengineering

Berkeley, CA

May 2028

Technical Skills: Solidworks, Google Suite, Canva, Microsoft Office, FEA Simulations, Waterjet Cutting, CNC Machine Operation, Soldering, Laser Cutting, Welding, Additive Manufacturing

Awards: Regents' and Chancellor's Scholarship, Yardi Scholarship, Arroyo High School Valedictorian, Horizon Hydrogen Grand Prix Nationals 5th Place, Kiwanis College Scholarship

Relevant Courses: Intro to Programming for Engineers, Three Dimensional Modeling for Design, Solid Mechanics

PROJECTS

Gen XI Solar Vehicle Pedal Box

January 2025 - Present

- Engineering pedals, mounting system, master cylinder housing and pedal arms of a vehicle using Solidworks; analyzing structures through FEA simulations
- 3D printing rapid prototypes of design and manufacturing all parts using CNC, Waterjet, and composite layups
- Delegating tasks for a sub-team of students, establishing deadlines, facilitating collaboration between other project teams

Gen XI Solar Vehicle Shell Mold

November 2024 - Present

- Designed the aerodynamic shell mold in Solidworks for composite manufacturing
- Welding a skid to support the shell mold during curing process

Wheel Rim Carrier CAD

September 2024

- Designed a wheel rim carrier on Solidworks to withstand a bump and brake force of a 700lb car
- Optimized geometry and weight through comprehensive FEA simulations to analyze the carrier's performance
- Presented professional design review including manufacturing and integration plans

PROFESSIONAL EXPERIENCE

Lawrence Berkeley National Laboratory

Engineering Student Assistant

Berkeley, CA

June 2025 - Present

- Working on the Advanced Light Source Upgrade project to remove existing x-ray infrastructure and installing new technology founded by recent accelerator physics breakthroughs
- Restructuring CAD files to BIM models for efficient contractor communication
- Designing and manufacturing diagnostic cable termination protection for 10,000+ BPM cables

East Bay Municipal Utilities District

Civil Engineering Intern

Oakland, CA

June 2023 - July 2023

- Collaborated with civil engineers in construction worksites, environmental consultations, and topographic surveys
- Wrote formal proposals to the Board of Directors encouraging for the continuation of the internship program
- Designed educational graphics on the water treatment process for public outreach

Tssui Research and Design

Materials Research Intern

Emeryville, CA

July 2023 - October 2023

- Researched properties of sustainable materials to be used in architectural projects in California and Shenzhen, China
- Presented research to a panel of employers

City of San Leandro

Transportation Department Intern

Oakland, CA

July 2022 - August 2022

- Designed and pitched new bike lane design to encourage safety, exercise and eco-friendly transportation routes for schools
- Delivered flyers to 800+ households and coordinated community events to spread awareness of upcoming bike lane and gather public feedback to refine development plans

EXTRACURRICULARS

Regents and Chancellor's Scholar Association (RCSA)

Corporate Relations Coordinator

Berkeley, CA

October 2024 - Present

- Coordinating, organizing and hosting guest speakers, career fairs and other educational events with industry professionals
- Planning and leading career preparation workshops

UC Berkeley Solar Vehicle Team (CalSol)

Dynamics Project Lead

Berkeley, CA

October 2024 - Present

- Defining project timelines in a sub-team working on a 150+ student project and presenting weekly findings to project managers